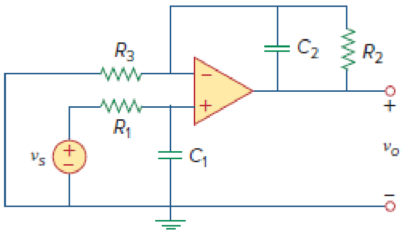
Chapter 10, Problem 77P

Problem

Compute the closed-loop gain **V***o*/**V***s* for the op amp circuit of Fig. 10.120.

Figure 10.120:



Step-by-step solution

Step 1 of 3

Refer to circuit diagram in figure 10.120 in the textbook.

Convert the circuit to the frequency domain.

$$C_1 \Rightarrow \frac{1}{j\omega C_1}$$

$$C_2 \Rightarrow \frac{1}{j\omega C_2}$$

Frequency domain equivalent circuit is,

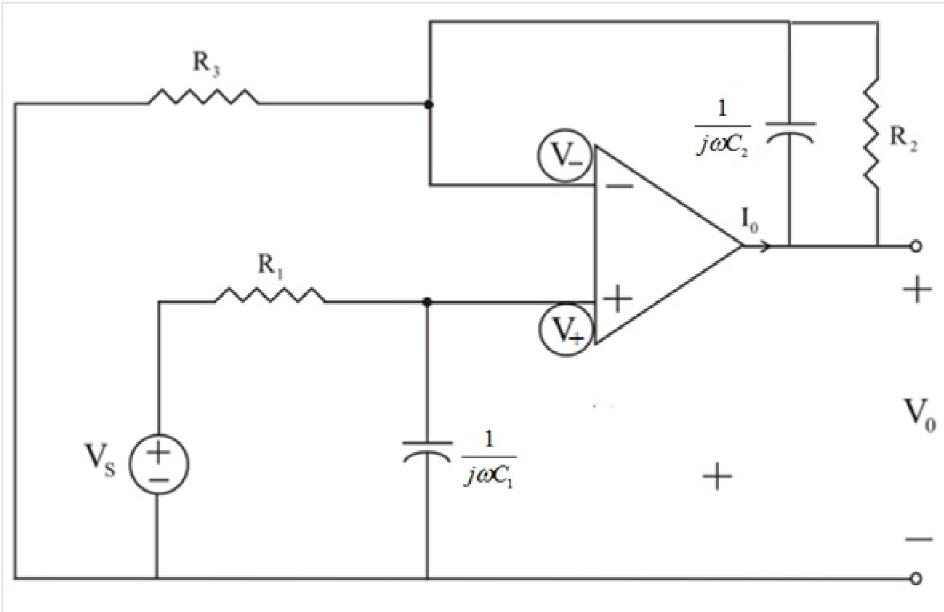


Figure 1

The voltage at non inverting terminal is equal to voltage at inverting terminal.

So, $v_+ = v_-$

Step 2 of 3

Apply Kirchhoff's current law (KCL) at non-inverting terminal.

$$\frac{v_+ - v_s}{R_1} + \frac{v_+}{\frac{1}{j\omega C_1}} = 0$$

$$\frac{v_+ - v_s}{R_1} + v_+\left(j\omega C_1\right) = 0$$

Simplify further.

$$v_+\left(\frac{1}{R_1} + j\omega C_1\right) - \frac{v_s}{R_1} = 0$$

$$v_+\left(\frac{1 + j\omega R_1 C_1}{R_1}\right) = \frac{v_s}{R_1}$$

$$v_+\left(1 + j\omega R_1 C_1\right) = v_s$$

$$v_+ = v_s\left(\frac{1}{1 + j\omega R_1 C_1}\right)$$

$$v_- = v_s\left(\frac{1}{1 + j\omega R_1 C_1}\right) \quad \text{since } v_+ = v_- \quad \text{......} \ (1)$$

Step 3 of 3

Apply Kirchhoff's current law (KCL) at inverting terminal.

$$\frac{v_-}{R_3} + \frac{v_- - v_o}{\frac{1}{j\omega C_2}} + \frac{v_- - v_o}{R_2} = 0$$

$$v_-\left(\frac{1}{R_3} + j\omega C_2 + \frac{1}{R_2}\right) = v_o\left(j\omega C_2 + \frac{1}{R_2}\right)$$

$$v_-\left(\frac{R_2 + j\omega R_2 R_3 C_2 + R_3}{R_3 R_2}\right) = v_o\left(\frac{j\omega R_2 C_2 + 1}{R_2}\right)$$

$$v_-\left(\frac{R_2 + j\omega R_2 R_3 C_2 + R_3}{R_3}\right) = v_o\left(j\omega R_2 C_2 + 1\right)$$

$$v_- = v_o\left(\frac{j\omega R_2 R_3 C_2 + R_3}{R_2 + j\omega R_2 R_3 C_2 + R_3}\right)$$

From equation (1), substitute $v_s\left(\frac{1}{1 + j\omega R_1 C_1}\right)$ for v_- in the equation.

$$v_s\left(\frac{1}{1 + j\omega R_1 C_1}\right) = v_o\left(\frac{j\omega R_2 R_3 C_2 + R_3}{R_2 + j\omega R_2 R_3 C_2 + R_3}\right)$$

$$v_s\left(\frac{R_2 + j\omega R_2 R_3 C_2 + R_3}{\left(1 + j\omega R_1 C_1\right)\left(j\omega R_2 R_3 C_2 + R_3\right)}\right) = v_o$$

Rearrange the terms.

$$\frac{v_o}{v_s} = \frac{R_2 + R_3 + j\omega C_2 R_2 R_3}{\left(1 + j\omega R_1 C_1\right)\left(R_3 + j\omega C_2 R_2 R_3\right)}$$

Thus, closed loop gain $\frac{v_o}{v_s}$ of the circuit is $\boxed{\frac{R_2 + R_3 + j\omega C_2 R_2 R_3}{\left(1 + j\omega R_1 C_1\right)\left(R_3 + j\omega C_2 R_2 R_3\right)}}$.